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SUMMARY

Computer science student at Purdue working across software engineering, ML, and human-centered research. Researcher in 3 labs (wearable sensing, robotics, XR security) with industry experience shipping production ML pipelines; leads 20-contributor software teams.

EDUCATION

Purdue University — College of Science <i>B.S. in Computer Science (Machine Intelligence & Systems Software Concentration)</i> • Minor: Mathematics • Relevant Coursework: Artificial Intelligence, Algorithm Analysis, Data Structures & Algorithms, Systems Programming (C, Assembly), Computer Architecture, Object-Oriented Programming (Java), Competitive Programming (C++), Linear Algebra, Multivariable Calculus	West Lafayette, IN Aug 2024 – May 2027
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RESEARCH EXPERIENCE

Purdue University — Snowball, CoRAL, and PurSec Labs <i>Undergraduate Researcher</i> • Snowball (Multimodal Wearable Sensing): Developed the processing pipeline for smart-earring sensor data (PPG, accelerometer, temperature, audio) — synchronization, cleaning, and dataset generation across 15+ participants / 30+ hours; engineered features and trained models for eating detection and chewing-count estimation (97% F1, 0.634 MAE). • CoRAL (Robot Learning): Analyzed state-of-the-art deep learning and Vision-Language Model (VLM) architectures for long-horizon robotic manipulation; conducted literature reviews and evaluated VLM reasoning on object-relocation tasks with obstacles across 10 scenarios. • PurSec (XR Security & Privacy): Conducted literature reviews on XR privacy and security, LLM-enabled AR attack vectors, and user/bystander risks. Designed a comprehensive survey and interview study (200+ participants) investigating awareness gaps, willingness to adopt defenses, and novel XR threats; reviewed 20+ papers.	West Lafayette, IN Nov 2024 – Present
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INDUSTRY EXPERIENCE

Reclub — Sports Technology Startup <i>Software Engineer Intern</i> • Designed and deployed an automated video segmentation system identifying pickleball rallies from raw match footage, reducing manual editing time from hours to ~5 minutes per match. • Built and optimized a YOLOv10s-based, motion-aware classification pipeline achieving 96% accuracy while targeting near real-time, on-device inference. • Implemented an end-to-end ML pipeline — 36K+ frame data labeling, training, and cloud deployment — using Docker, Google Cloud Storage, and Vertex AI. • Productionized model inference workflows supporting scalable content generation, contributing to increased user engagement and platform adoption.	Ho Chi Minh City, VN May 2025 – Aug 2025
Amazon Web Services (AWS) Youth Experience Program <i>Data Analytics Intern</i> • Developed Python-based data processing pipelines (Pandas, NumPy, scikit-learn) to analyze wildfire risk, predicting 7 forest cover types from 54 cartographic features on the 581K-sample UCI Covertypes dataset. • Evaluated 5 model families (Logistic Regression, Decision Trees, Random Forests, Neural Networks, KNN) to identify optimal features and improve predictive accuracy; recognized as a top-5 intern project.	Remote Jul 2023 – Aug 2023
BuyMed — \$64M Healthcare Technology Platform Startup <i>Retail Operations Intern</i> • Conducted market research on competitors and the healthcare landscape; analyzed operational data across 3 pharmacy partners and authored standard operating procedures to improve efficiency.	Ho Chi Minh City, VN Jul 2023 – Aug 2023

UNIVERSITY EXPERIENCE

Purdue University — College of Liberal Arts <i>Web Developer</i> • Maintain and extend university web services by developing reusable templates and components using HTML, CSS, JavaScript, and Apache Velocity. • Resolve support tickets involving content updates, accessibility improvements, and legacy system maintenance.	West Lafayette, IN Jan 2026 – Present
Purdue University <i>COSINE Mathematics Tutor</i> • Supported 50+ undergraduates across Precalculus and Calculus I–II, adapting instruction to diverse learning styles.	West Lafayette, IN Jan 2025 – May 2025

LEADERSHIP & TEACHING

Purdue ACM SIGAPP

President & Technical Lead

West Lafayette, IN

Aug 2025 – Present

- Led backend development for two large-scale student-run software projects with ~20 contributors each, coordinating architecture decisions, sprint planning, and code reviews.
- Designed REST APIs, database schemas, and authentication flows for *BoilerBuzz*, a cross-platform mobile app backed by Node.js and PostgreSQL.
- Architected a FastAPI-based grant management platform for Purdue PGSG supporting submission, double-blind review, approval, and reimbursement workflows; implemented role-based access control with Keycloak for secure multi-role, multi-stage approval pipelines.

SSIS Programming Club

Co-Leader

Ho Chi Minh City, VN

Aug 2023 – May 2024

- Taught 15 students data structures, trees, and algorithms in C++ and Python to prepare for USACO competitions; led weekly timed practice sessions and facilitated student presentations on advanced CS topics.

Schoolhouse.world — Free Peer Tutoring Platform

Volunteer Peer Tutor (SAT Prep & Enrichment)

Remote

May 2023 – Aug 2023

- Delivered 75+ hours of free tutoring across 56 sessions to 72 learners in 17 countries; ranked top 5% of tutors platform-wide in hours, sessions, learner reach, and positive ratings (165+).

Unity Impact — SSIS Video Game Development Club

Co-founder & President

Ho Chi Minh City, VN

Aug 2022 – Dec 2023

- Designed curriculum, organized a public game exhibition (30+ visitors), and led a winter bootcamp for 21 participants from Vietnam, Australia, and the United States.
- Guided development of 10 games across puzzle to adventure genres using Unity and C#, incorporating 2D sprites and 3D models.

PROJECTS

Boiler-Gram

Multi-threaded Java Client-Server Application

West Lafayette, IN

Oct 2024 – Dec 2024

- Built a multi-threaded client-server system with authentication, messaging, and persistent storage; comprehensive JUnit test coverage.

Roadie Rescue — 3rd Place of 35 Teams, Hello World Hackathon (Devpost / GitHub)

Real-Time Roadside Assistance Platform (MERN)

West Lafayette, IN

Oct 2024

- Developed a real-time emergency assistance platform with location-based requests and live status updates.

Site Scouter — Catapult Hackathon (Devpost / Demo / GitHub)

Geospatial Renewable-Energy Site-Selection Platform

West Lafayette, IN

Apr 2026

- Built a full-stack tool scoring user-drawn regions for solar/wind/data-center siting from satellite, terrain, and OpenStreetMap data; PyTorch model trained on thousands of real projects predicts power generation with $R^2 > 0.90$ (FastAPI, React, Leaflet, AWS).

Talking Slides — BoilerMake Hackathon (Devpost / GitHub)

PDF-to-Podcast Study Tool

West Lafayette, IN

Jan 2025

- Converted PDFs into multi-voice podcast audio via the OpenAI API — Django REST, PyMuPDF, MongoDB/GridFS, Redis caching, Auth0-secured React frontend.

Unity WebGL Games — Hexagon World (GitHub) & Save The Chicken (GitHub)

Procedural Generation & Arcade Games (Unity, C#)

Browser-Playable

2022

- Shipped two browser-playable games: a procedural hexagonal-grid world simulation and a physics-based reflex arcade game, both exported to WebGL and hosted on GitHub Pages.

SchoolOfAthens.world

Platform for Youth Debate Perspectives (Next.js, Firebase)

Remote

Jan 2023 – Jun 2023

- Built a real-time social debate platform for youths with 7 international developers.

SKILLS

- **Languages:** Python, Java, C++, C, JavaScript, TypeScript, SQL, Assembly
- **Backend & Systems:** FastAPI, Django, Node.js, Express, REST APIs, PostgreSQL, MongoDB, Redis, Keycloak (RBAC), multithreading
- **ML & Data:** PyTorch, scikit-learn, Pandas, NumPy, YOLO/object detection, LLM/OpenAI API integration, feature engineering, model evaluation, sensor signal processing, geospatial analysis
- **Cloud & DevOps:** Docker, Google Cloud Platform (GCS, Vertex AI), Git/GitHub
- **Game Development:** Unity, C#, procedural generation, WebGL builds, 2D/3D asset integration

HONORS & AWARDS

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- 3rd Place, Hello World Hackathon, Purdue (2024); Top-5 Intern Project, AWS Youth Experience Program (2023)
 - 2nd Place, Girls In Tech Code/No-Code Hack (2022); Most Voted Innovative Idea, Poolesville HS Hackathon (2022)
 - 1st Place, SSIS Inclusive Invention Challenge (2020)